OPERATIONS / FLIGHT DELAY ANALYSIS

Turning flight operations data into actionable business insights.

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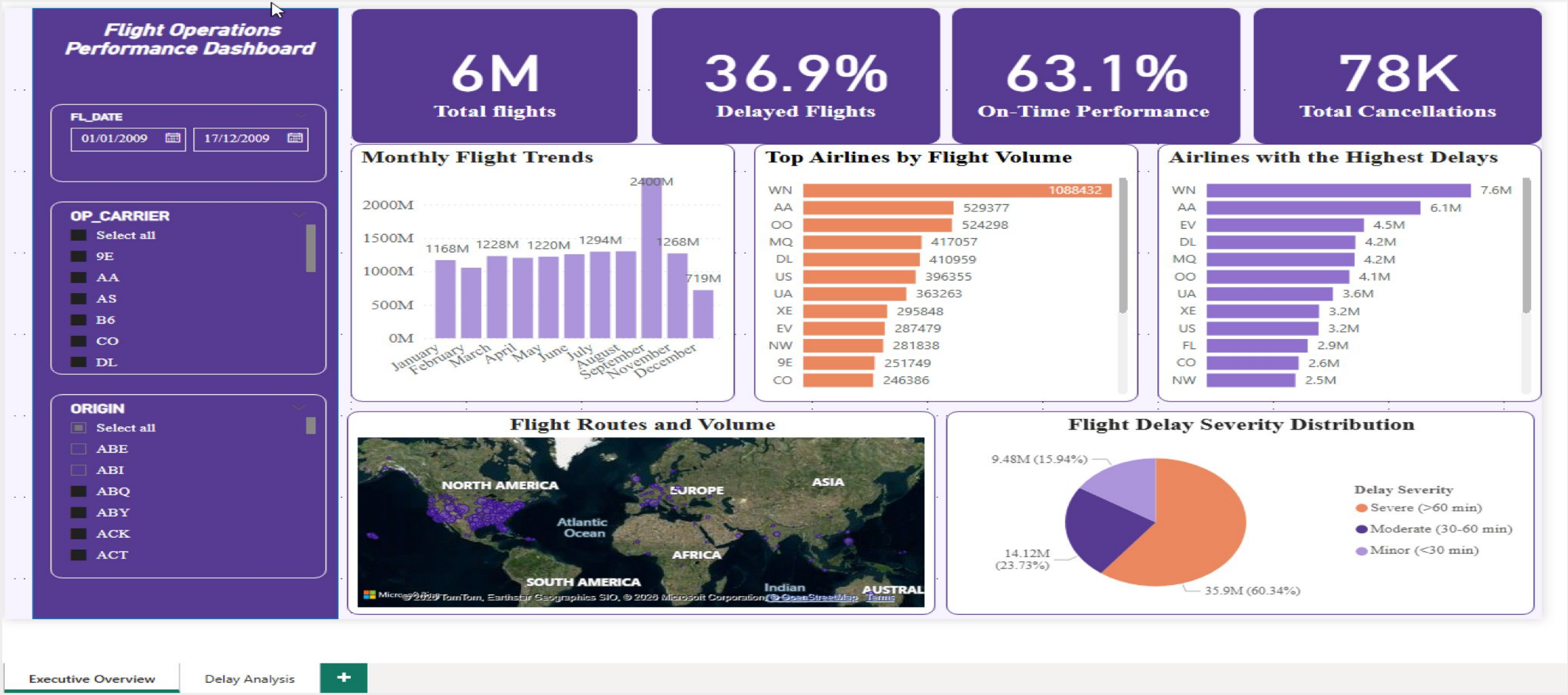
Dashboard Overview



A high-level view of key flight operations performance indicators, providing a quick snapshot of overall delays, cancellations and airline performance.

Introduction

Airline delays can have a significant impact on operational efficiency, passenger experience and overall airline performance.

This project analyzes flight delay data to identify patterns in:

- Delay volume
- Airline performance
- Monthly delay trends
- Delay severity
- Airport-level performance
- Flight cancellations

Project Objective

To analyze flight delay patterns and identify areas where airline operations could be improved through data-driven decision-making.

Background

Flight delays can occur for different operational and external reasons, including:

- Carrier-related issues
- Weather conditions
- National Airspace System (NAS) delays
- Security-related delays
- Late-arriving aircraft

The dataset contains separate fields for these delay categories rather than a single "Delay Type" field.

Business Questions

The analysis was designed to answer:

1 How much delay is occurring?

2 How many flights are delayed?

3 Which airlines experience the highest number of delays?

4 Which airlines have the highest average delay?

5 When are delays highest?

6 Where are delays occurring?

7 How severe are the delays?

These questions directly informed the dashboard design.

Methodology

Data Preparation

- Imported flight operations data into Power BI
- Reviewed and prepared the dataset using Power Query
- Checked relevant fields and data types
- Created calculated measures for analysis

Data Analysis

Created KPIs and measures to evaluate:

- Total Delay Minutes
- Average Delay Per Flight
- Delayed Flights
- Cancellation Rate
- Delay Severity

Visualization

Developed interactive Power BI visuals to analyze:

- Airline performance
- Monthly trends
- Delay severity
- Origin airport performance

Approach

Data → Preparation → Analysis → Visualization → Insights → Recommendations

Analysis

KPI	Result
Total Delay Minutes	63.52M
Average Delay Per Flight	9.88 min
Delayed Flights	1M
Cancellation Rate	1.4%

These KPIs provide an overview of the scale of delays and cancellations in the dataset.

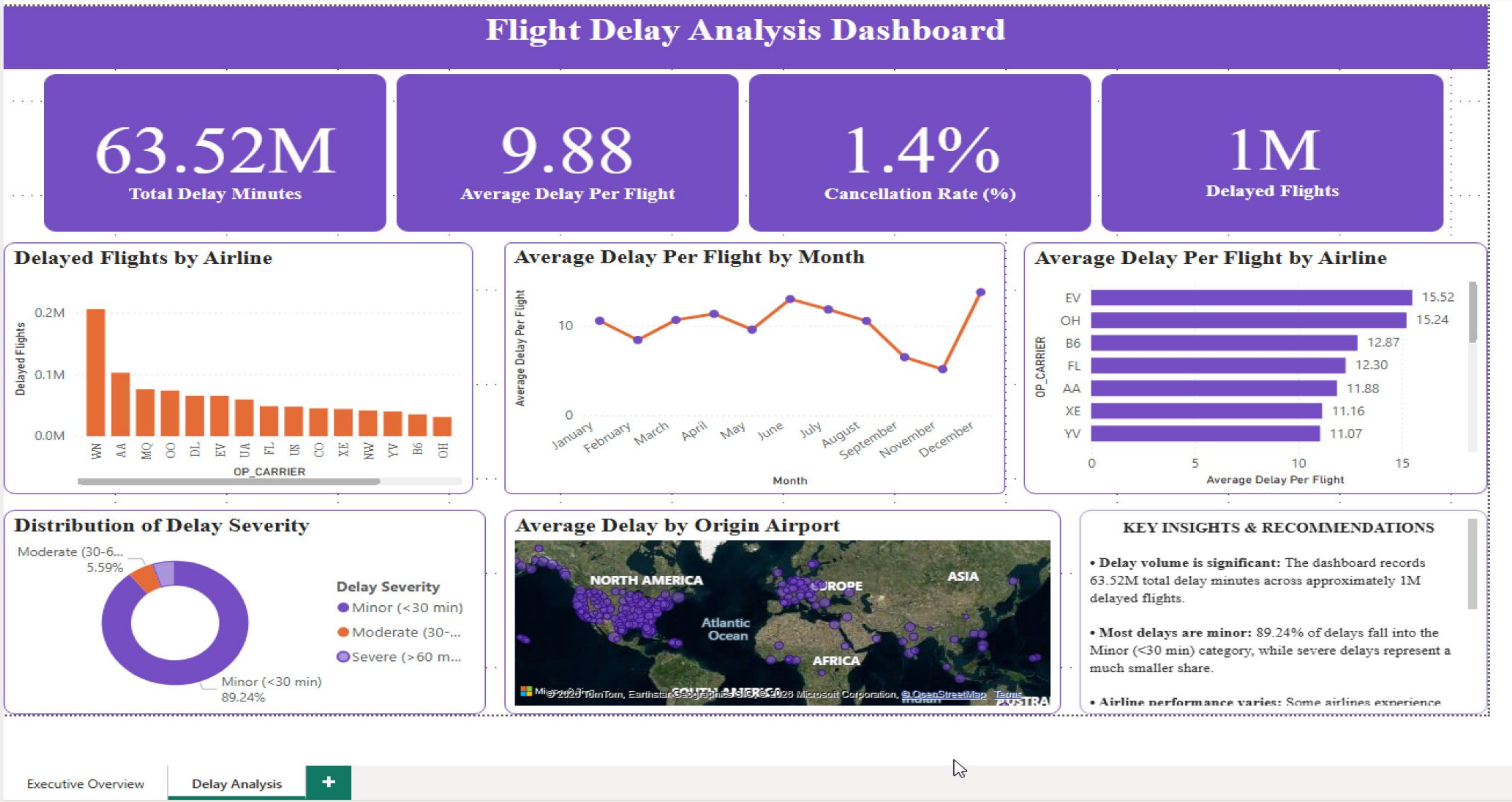
Airline Performance

The analysis uses two different perspectives:

Delayed Flights by Airline
→ Measures the **volume** of delayed flights.

Average Delay Per Flight by Airline
→ Measures the **average delay duration**.

This distinction is important because an airline can have many delayed flights without necessarily having the longest average delays.



Flight Delay Analysis - Airline performance, delay trends, severity and airport-level analysis

Analysis - Trends & Severity

Monthly Delay Trend

The **Average Delay Per Flight by Month** line chart was used to identify changes in delay levels throughout the year.

This helps answer: **When are delays higher or lower?**

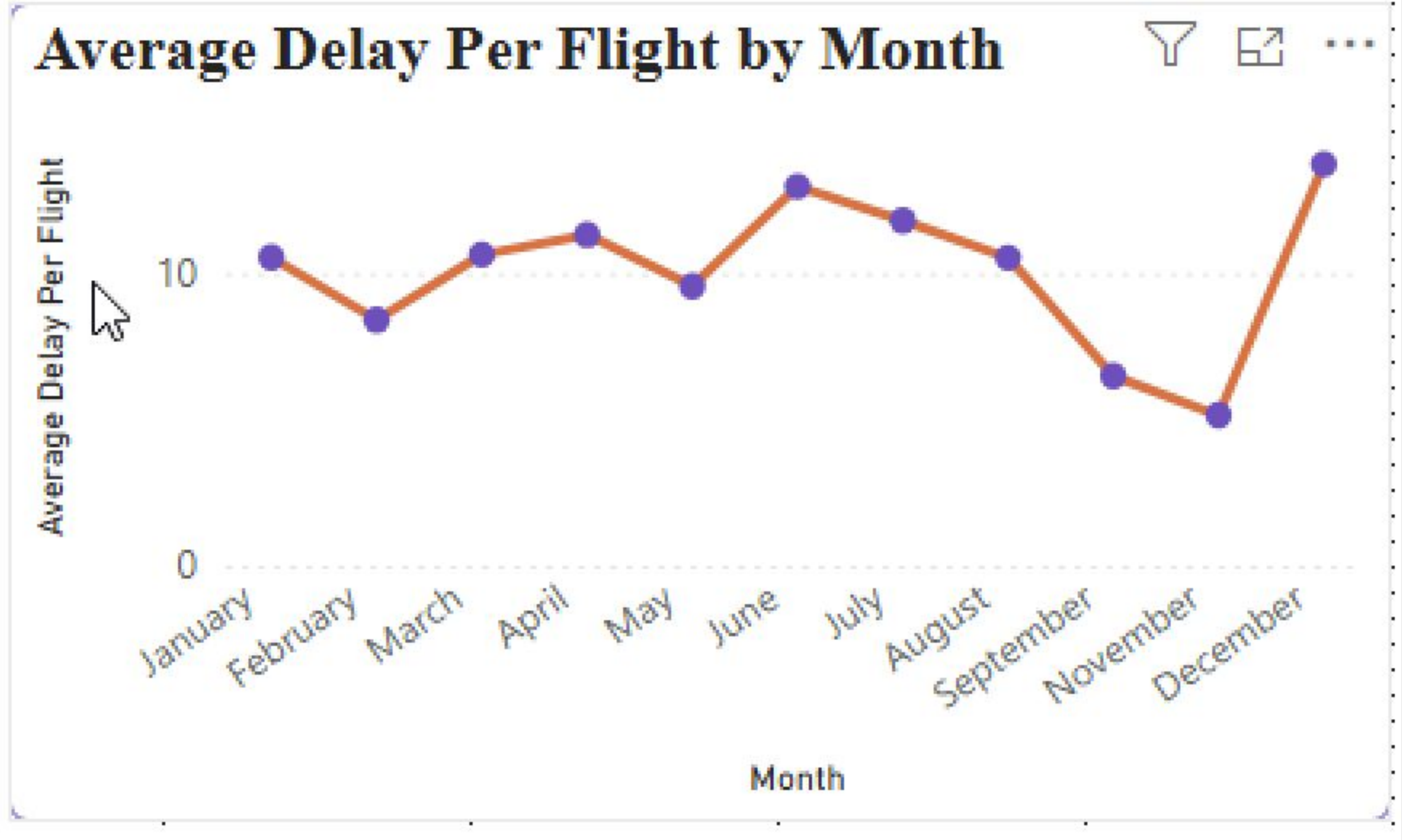
Delay Severity

The analysis classified delays into:

- **Minor:** <30 minutes
- **Moderate:** 30–60 minutes
- **Severe:** >60 minutes

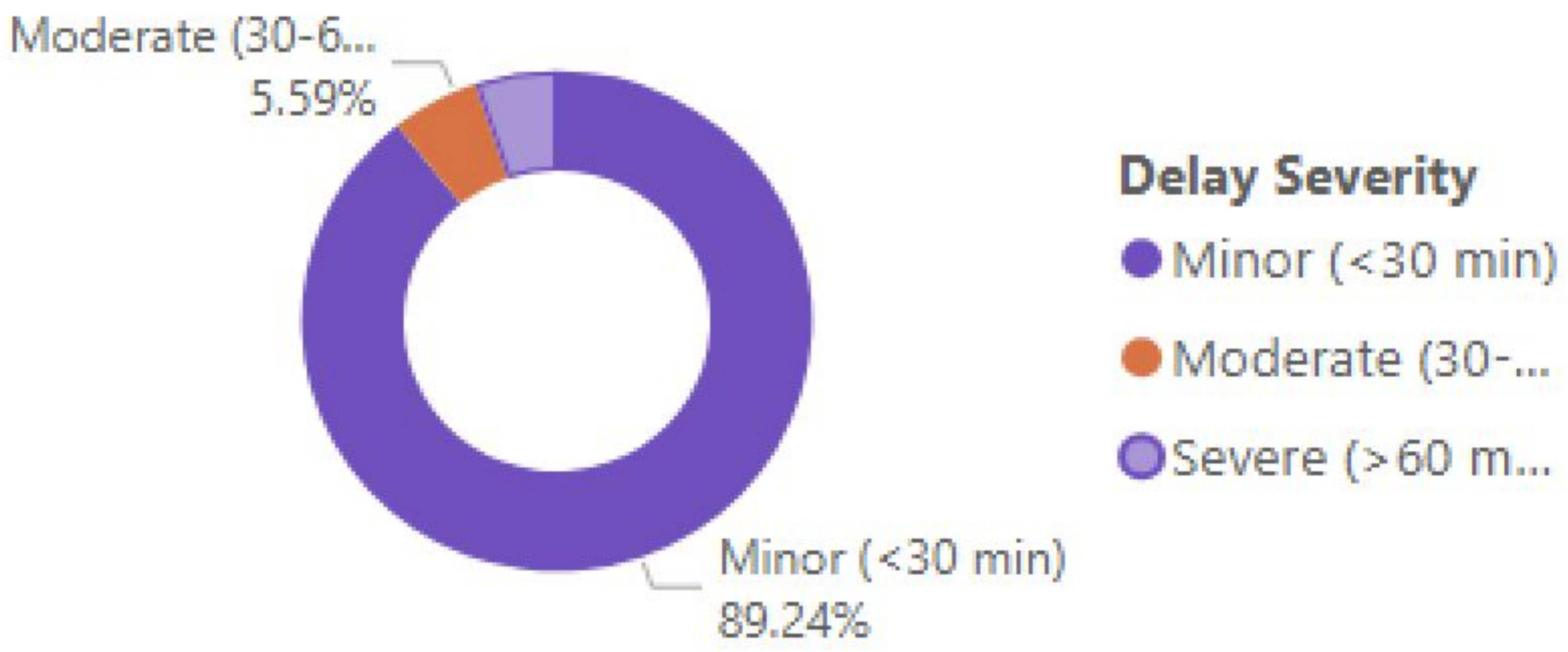
The dashboard shows that: **89.24% of delays are Minor**

while **5.59% are Moderate**, with the remaining proportion classified as Severe.


Average Delay Per Flight by Month
Average Delay Per Flight
10
0
January
February
March
April
May
June
July
August
September
November
December
Month


Average Delay Per Flight by Airline
OP_CARRIER
EV
OH
B6
FL
AA
XE
YV
15.52
15.24
12.87
12.30
11.88
11.16
11.07
0
5
10
15
Average Delay Per Flight

Distribution of Delay Severity



Key Insights

Delays represent a significant operational issue

The dataset records approximately **63.52 million total delay minutes** across approximately **1 million delayed flights**.

Most delays are relatively short

89.24% of delays fall into the Minor category (<30 minutes), indicating that most delays are short-duration events.

Airline performance varies

The average delay analysis shows differences between airlines.

Among the airlines displayed, **EV recorded the highest average delay at 15.52 minutes**, followed by **OH at 15.24 minutes**.

Delays vary over time

The monthly analysis shows that delay performance changes throughout the year, highlighting periods that may require closer operational monitoring.

Delay patterns differ by location

The airport map provides a geographical view of average delay performance and helps identify where delays are occurring.

Recommendations

Based on the analysis, airlines and airport operators could:

1. Prioritize high-delay airlines

Investigate airlines consistently recording higher average delays and identify operational bottlenecks.

2. Monitor high-risk periods

Use monthly delay trends to anticipate periods of increased delays and allocate resources proactively.

3. Investigate airport-level patterns

Identify airports with consistently high average delays and investigate local operational factors.

4. Focus on reducing recurring minor delays

Although most delays are minor, their high frequency can accumulate into a substantial number of lost operational minutes.

5. Use data for continuous monitoring

Establish regular KPI monitoring for delay volume, average delay, cancellations and airline performance.

Reducing the frequency and duration of recurring delays can improve operational efficiency and passenger experience.

Conclusion

This project demonstrates how Power BI can be used to transform flight operations data into a clear and actionable business intelligence solution.

The analysis moved from:

Raw Data → Data Preparation → KPI Analysis → Visualization → Business Insights → Recommendations

Key Value Delivered

- Identified the scale of flight delays
- Compared airline performance
- Analyzed monthly delay patterns
- Examined delay severity
- Highlighted airport-level patterns
- Converted findings into actionable recommendations

Tools Used

Power BI | Power Query | DAX | Data Visualization | Business Analysis